

3-D geometry

1. If vertices of a triangle are $A(1, -1, 2), B(2, 0, -1)$ and $C(0, 2, 1)$, then the area of a triangle is
 - (a) $\sqrt{6}$
 - (b) $2\sqrt{6}$
 - (c) $3\sqrt{6}$
 - (d) $4\sqrt{6}$
2. The xy -plane divides the line joining the points $(-1, 3, 4)$ and $(2, -5, 6)$
 - (a) Internally in the ratio $2 : 3$
 - (b) Internally in the ratio $3 : 2$
 - (c) Externally in the ratio $2 : 3$
 - (d) Externally in the ratio $3 : 2$
3. The ratio in which the line joining $(2, 4, 5)$ and $(3, 5, -4)$ is divided by the yz -plane is
 - (a) $2 : 3$
 - (b) $3 : 2$
 - (c) $-2 : 3$
 - (d) $4 : -3$
4. The angle between the straight lines $\frac{x+1}{2} = \frac{y-2}{5} = \frac{z+3}{4}$ and $\frac{x-1}{1} = \frac{y+2}{2} = \frac{z-3}{-3}$ is
 - (a) 45°
 - (b) 30°
 - (c) 60°
 - (d) 90°
5. If direction cosines of two lines are proportional to $(2, 3, -6)$ and $(3, -4, 5)$, then the acute angle between them is
 - (a) $\cos^{-1}\left(\frac{49}{36}\right)$
 - (b) $\cos^{-1}\left(\frac{18\sqrt{2}}{35}\right)$
 - (c) 90°
 - (d) $\cos^{-1}\left(\frac{18}{35}\right)$
6. If $P(3, 4, 5), Q(4, 6, 3), R(-1, 2, 4), S(1, 0, 5)$ then the projection of RS on PQ is
 - (a) $-2/3$
 - (b) $-4/3$
 - (c) $1/2$
 - (d) 2
7. The equation to the straight line passing through the points $(4, -5, -2)$ and $(-1, 5, 3)$ is
 - (a) $\frac{x-4}{1} = \frac{y+5}{-2} = \frac{z+2}{-1}$
 - (b) $\frac{x+1}{1} = \frac{y-5}{2} = \frac{z-3}{-1}$
 - (c) $\frac{x}{-1} = \frac{y}{5} = \frac{z}{3}$
 - (d) $\frac{x}{4} = \frac{y}{-5} = \frac{z}{-2}$
8. The shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is
 - (a) $\sqrt{30}$
 - (b) $2\sqrt{30}$
 - (c) $5\sqrt{30}$
 - (d) $3\sqrt{30}$
9. The equation of a plane parallel to x -axis is
 - (a) $ax + by + cz + d = 0$
 - (b) $ax + by + d = 0$
 - (c) $by + cz + d = 0$
 - (d) $ax + cz + d = 0$
10. The plane $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$ cuts the axes in A, B, C , then the area of the $\triangle ABC$ is
 - (a) $\sqrt{29}$
 - (b) $\sqrt{41}$
 - (c) $\sqrt{61}$
 - (d) None of these
11. The equation of the plane through the point $(1, 2, 3)$ and parallel to the plane $x + 2y + 5z = 0$ is
 - (a) $(x-1) + 2(y-2) + 5(z-3) = 0$
 - (b) $x + 2y + 5z = 14$
 - (c) $x + 2y + 5z = 6$
 - (d) None of these
12. The equation of the plane passing through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $4x + 3y + 2z + 1 = 0$ and the origin is
 - (a) $3x + 2y + z + 1 = 0$
 - (b) $3x + 2y + z = 0$
 - (c) $2x + 3y + z = 0$
 - (d) $x + y + z = 0$
13. A plane which passes through the point $(3, 2, 0)$ and the line $\frac{x-3}{1} = \frac{y-6}{5} = \frac{z-4}{4}$ is
 - (a) $x - y + z = 1$
 - (b) $x + y + z = 5$
 - (c) $x + 2y - z = 0$
 - (d) $2x - y + z = 5$
14. The line joining the points $(3, 5, -7)$ and $(-2, 1, 8)$ meets the yz -plane at point
 - (a) $\left(0, \frac{13}{5}, 2\right)$
 - (b) $\left(2, 0, \frac{13}{5}\right)$
 - (c) $\left(0, 2, \frac{13}{5}\right)$
 - (d) $(2, 2, 0)$

15. The equation of the plane through the intersection of the planes $x + y + z = 1$ and $2x + 3y - z + 4 = 0$ and parallel to x -axis is
- (a) $y - 3z + 6 = 0$ (b) $3y - z + 6 = 0$
 (c) $y + 3z + 6 = 0$ (d) $3y - 2z + 6 = 0$
16. The value of k such that $\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$ lies in the plane $2x - 4y + z = 7$, is
- (a) 7 (b) -7
 (c) No real value (d) 4
17. The length of the perpendicular from the origin to the plane $3x + 4y + 12z = 52$ is
- (a) 3 (b) -4
 (c) 5 (d) None of these
18. The value of $aa' + bb' + cc'$ being negative the origin will lie in the acute angle between the planes $ax + by + cz + d = 0$ and $a'x + b'y + c'z + d' = 0$, if
- (a) $a = a' = 0$
 (b) d and d' are of same sign
 (c) d and d' are of opposite sign
 (d) None of these
19. The angle between the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and the plane $3x + 2y - 3z = 4$ is
- (a) 45° (b) 0°
 (c) $\cos^{-1}\left(\frac{24}{\sqrt{29}\sqrt{22}}\right)$ (d) 90°
20. Let $(3, 4, -1)$ and $(-1, 2, 3)$ are the end points of a diameter of sphere. Then the radius of the sphere is equal to
- (a) 1 (b) 2
 (c) 3 (d) 9
21. The lines $x = ay + b$, $z = cy + d$ and $x = a'y + b'$, $z = c'y + d'$ are perpendicular to each other, if
- (a) $aa' + cc' = 1$ (b) $aa' + cc' = -1$
 (c) $ac + a'c' = 1$ (d) $ac + a'c' = -1$
22. If a line makes the angle α, β, γ with three dimensional coordinate axes respectively, then $\cos 2\alpha + \cos 2\beta + \cos 2\gamma =$
- (a) -2 (b) -1
- (c) 1 (d) 2
23. The angle between the planes $2x - y + z = 6$ and $x + y + 2z = 7$ is
- (a) 30° (b) 45°
 (c) 0° (d) 60°
24. The distance of the point $(-1, -5, -10)$ from the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 5$, is
- (a) 10 (b) 11
 (c) 12 (d) 13
25. From which of the following the distance of the point $(1, 2, 3)$ is $\sqrt{10}$
- (a) Origin (b) x -axis
 (c) y -axis (d) z -axis
26. If α, β, γ be the angles which a line makes with the positive direction of co-ordinate axes, then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma =$
- [RPET 2000; AMU 2002; MP PET 1989, 98, 2000, 03, Pb. CET 2001]
- (a) 2 (b) 1
 (c) 3 (d) 0
27. If the projections of a line on the co-ordinate axes be $2, -1, 2$, then the length of the line is
- (a) 3 (b) 4
 (c) 2 (d) $\frac{1}{2}$
28. The co-ordinates of a point which is equidistant from the points $(0, 0, 0), (a, 0, 0), (0, b, 0)$ and $(0, 0, c)$ are given by
- [MP PET 1993]
- (a) $\left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$ (b) $\left(-\frac{a}{2}, -\frac{b}{2}, \frac{c}{2}\right)$
 (c) $\left(\frac{a}{2}, -\frac{b}{2}, -\frac{c}{2}\right)$ (d) $\left(-\frac{a}{2}, \frac{b}{2}, -\frac{c}{2}\right)$
29. If l_1, m_1, n_1 and l_2, m_2, n_2 are the direction cosines of two perpendicular lines, then the direction cosine of the line which is perpendicular to both the lines, will be
- (a) $(m_1n_2 - m_2n_1), (n_1l_2 - n_2l_1), (l_1m_2 - l_2m_1)$
 (b) $(l_1l_2 - m_1m_2), (m_1m_2 - n_1n_2), (n_1n_2 - l_1l_2)$
 (c) $\frac{1}{\sqrt{l_1^2 + m_1^2 + n_1^2}}, \frac{1}{\sqrt{l_2^2 + m_2^2 + n_2^2}}, \frac{1}{\sqrt{3}}$
 (d) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$
30. If $\left(\frac{1}{2}, \frac{1}{3}, n\right)$ are the direction cosines of a line, then the value of n is
- [Kerala (Engg.) 2002]
- (a) $\frac{\sqrt{23}}{6}$ (b) $\frac{23}{6}$
 (c) $\frac{2}{3}$ (d) $\frac{3}{2}$
31. The line $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar, if
- [AIEEE 2003]
- (a) $k = 0$ or -1 (b) $k = 0$ or 1
 (c) $k = 0$ or -3 (d) $k = 3$ or -3

32. A line makes the same angle θ , with each of the x and z -axis. If the angle β , which it makes with y -axis is such that $\sin^2 \beta = 3 \sin^2 \theta$, then $\cos^2 \theta$ equals [AIEEE 2004]
- (a) $\frac{3}{5}$ (b) $\frac{2}{3}$
(c) $\frac{1}{5}$ (d) None of these
33. The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$, is [MP PET 1994, 99; AIEEE 2005]
- (a) 0° (b) 30°
(c) 45° (d) 90°
34. The ratio in which the plane $x - 2y + 3z = 17$ divides the line joining the points $(-2, 4, 7)$ and $(3, -5, 8)$ is [AISSE 1988]
- (a) $10 : 3$ (b) $3 : 1$
(c) $3 : 10$ (d) $10 : 1$
35. The plane $ax + by + cz = 1$ meets the co-ordinate axes in A, B and C . The centroid of the triangle is [CET 1992]
- (a) $(3a, 3b, 3c)$ (b) $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$
(c) $\left(\frac{3}{a}, \frac{3}{b}, \frac{3}{c}\right)$ (d) $\left(\frac{1}{3a}, \frac{1}{3b}, \frac{1}{3c}\right)$
36. A plane which passes through the point $(3, 2, 0)$ and the line $\frac{x-3}{1} = \frac{y-6}{5} = \frac{z-4}{4}$ is [AIEEE 2002]
- (a) $x - y + z = 1$ (b) $x + y + z = 5$
(c) $x + 2y - z = 0$ (d) $2x - y + z = 5$
37. The angle between the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and the plane $3x + 2y - 3z = 4$ is [MP PET 2003]
- (a) 45° (b) 0°
(c) $\cos^{-1}\left(\frac{24}{\sqrt{29}\sqrt{22}}\right)$ (d) 90°
38. Equation $ax^2 + by^2 + cz^2 + 2fyz + 2gxz + 2hxy + 2ux + 2vy + 2wz + d = 0$ represents a sphere, if [MP PET 1990]
- (a) $a = b = c$
(b) $f = g = h = 0$
(c) $v = u = w$
(d) $a = b = c$ and $f = g = h = 0$
39. The centre of sphere passes through four points $(0, 0, 0)$, $(0, 2, 0)$, $(1, 0, 0)$ and $(0, 0, 4)$ is [MP PET 2002]
- (a) $\left(\frac{1}{2}, 1, 2\right)$ (b) $\left(-\frac{1}{2}, 1, 2\right)$
(c) $\left(\frac{1}{2}, 1, -2\right)$ (d) $\left(1, \frac{1}{2}, 2\right)$
40. If the plane $2ax - 3ay + 4az + 6 = 0$ passes through the midpoint of the line joining the centres of the spheres $x^2 + y^2 + z^2 + 6x - 8y - 2z = 13$ and $x^2 + y^2 + z^2 - 10x + 4y - 2z = 8$, then a equals [AIEEE 2005]
- (a) -2 (b) 2
(c) -1 (d) 1
41. The plane $x + 2y - z = 4$ cuts the sphere $x^2 + y^2 + z^2 - x + z - 2 = 0$ in a circle of radius [AIEEE 2005]
- (a) 2 (b) $\sqrt{2}$
(c) 3 (d) 1
42. The radius of sphere $x + 2y + 2z = 15$ and $x^2 + y^2 + z^2 - 2y - 4z = 11$ is [AMU 2005]
- (a) 2 (b) $\sqrt{7}$
(c) 3 (d) $\sqrt{5}$
43. The equation of the sphere concentric with the sphere $2x^2 + 2y^2 + 2z^2 - 6x + 2y - 4z = 1$ and double its radius is [Kerala (Engg.) 2005]
- (a) $x^2 + y^2 + z^2 - x + y - z = 1$
(b) $x^2 + y^2 + z^2 - 6x + 2y - 4z = 1$
(c) $2x^2 + 2y^2 + 2z^2 - 6x + 2y - 4z - 15 = 0$
(d) $x^2 + y^2 + z^2 - 3x + y - 2z = 1$
(e) $2x^2 + 2y^2 + 2z^2 - 6x + 2y - 4z - 25 = 0$
44. If $(2, 3, 5)$ is one end of a diameter of the sphere $x^2 + y^2 + z^2 - 6x - 12y - 2z + 20 = 0$ then co-ordinates of the other end of the diameter are [Kerala (Engg.) 2005]
- (a) $(4, 3, 5)$ (b) $(4, 9, -3)$
(c) $(4, 9, 3)$ (d) $(4, 3, -3)$
(e) $(4, 9, 5)$
45. The angle between two diagonals of a cube will be [MP PET 1996, 2000; RPET 2000, 02; UPSEAT 2004]
- (a) $\sin^{-1} 1/3$ (b) $\cos^{-1} 1/3$
(c) Variable (d) None of these
46. If the straight lines $x = 1 + s$, $y = -3 - \lambda s$, $z = 1 + \lambda s$ and $x = t/2$, $y = 1 + t$, $z = 2 - t$, with parameters s and t respectively, are co-planar, then λ equals [AIEEE 2004]
- (a) 0 (b) -1
(c) $-1/2$ (d) -2
47. The co-ordinates of the foot of perpendicular drawn from point $P(1, 0, 3)$ to the join of points $A(4, 7, 1)$ and $B(3, 5, 3)$ is [RPET 2001]
- (a) $(5, 7, 1)$ (b) $\left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3}\right)$
(c) $\left(\frac{2}{3}, \frac{5}{3}, \frac{7}{3}\right)$ (d) $\left(\frac{5}{3}, \frac{2}{3}, \frac{7}{3}\right)$
48. If the lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{1} = \frac{z}{1}$ intersect, then $k =$ [IIT Screening 2004]
- (a) $\frac{2}{9}$ (b) $\frac{9}{2}$
(c) 0 (d) None of these
49. A line with direction cosines proportional to $2, 1, 2$ meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. The co-ordinates of each of the points of intersection are given by [AIEEE 2004]
- (a) $(2a, a, 3a), (2a, a, a)$ (b) $(3a, 2a, 3a), (a, a, a)$
(c) $(3a, 2a, 3a), (a, a, 2a)$ (d) $(3a, 3a, 3a), (a, a, a)$
50. The equation of the plane passing through the points $(1, -3, -2)$ and perpendicular to planes $x + 2y + 2z = 5$ and $3x + 3y + 2z = 8$, is [AISSE 1987]
- (a) $2x - 4y + 3z - 8 = 0$ (b) $2x - 4y - 3z + 8 = 0$
(c) $2x + 4y + 3z + 8 = 0$ (d) None of these
51. A variable plane at a constant distance p from origin meets the co-ordinates axes in A, B, C . Through these points planes are drawn parallel to co-ordinate planes. Then locus of the point of intersection is
- (a) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2}$ (b) $x^2 + y^2 + z^2 = p^2$
(c) $x + y + z = p$ (d) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = p$
52. The d.r.'s of normal to the plane through $(1, 0, 0)$, $(0, 1, 0)$ which makes an angle $\frac{\pi}{4}$ with plane $x + y = 3$, are [AIEEE 2002]

Answer Sheet									
1	2	3	4	5	6	7	8	9	10
b	c	c	d	b	b	b	d	c	c
11	12	13	14	15	16	17	18	19	20
a	b	a	a	a	a	d	b	b	c
21	22	23	24	25	26	27	28	29	30
b	b	d	d	c	a	a	a	a	a
31	32	33	34	35	36	37	38	39	40
c	a	d	c	d	a	b	d	a	a
41	42	43	44	45	46	47	48	49	50
d	b	e	b	b	d	b	b	b	a
51	52	53	54	55	56	57	58	59	60
a	b	d	a	a	c	a	c	c	a
61									
b									